

Membrane Air Separation

Final Lab Report

Unit Operations Lab 5

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Group 4

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1. Executive Summary

By passing dry air through two hollow fiber polymeric membrane modules, the main objectives of this experiment were to determine the overall mass transport coefficients $O_2 \wedge N_2$ for a parallel and series configuration of the modules. Both configurations were performed at three air pressures: 20 psi, 40 psi, and 80 psi. For each air pressure, the bottom air flow rate was set at 0.5, 1.0, 2.0, 4.0, and 8.0 $\frac{L}{min}$ using volumetric flowmeters. As the air travels through the system, two digital O_2 analyzers were used to monitor the amount of O_2 as a percentage, in which the readings were allowed to stabilize before being recorded. A summary of the calculated values for each airflow rate, along with the corresponding air pressure and configuration, is shown below. Note: $K O_2$ and $K N_2$ represent the mass transfer coefficients of $O_2 \wedge N_2$, respectively.

Table 1: Summary of Species Mass Transfer Coefficients for Parallel Configuration

Parallel Configuration			
Air Pressure (psi)	Bottom Air Flow Rate ($\frac{L}{min}$)	$K O_2(\frac{mol}{min * Pa})$	$K N_2(\frac{mol}{min * Pa})$
20	0.5	1.42×10^{-4}	4.38×10^{-5}
	1.0	1.57×10^{-4}	4.42×10^{-5}
	2.0	1.72×10^{-4}	4.29×10^{-5}
	4.0	1.79×10^{-4}	4.12×10^{-5}
	8.0	1.96×10^{-4}	4.18×10^{-5}
40	0.5	1.66×10^{-4}	5.05×10^{-5}
	1.0	1.70×10^{-4}	5.04×10^{-5}
	2.0	1.80×10^{-4}	4.98×10^{-5}
	4.0	2.08×10^{-4}	4.89×10^{-5}
	8.0	2.35×10^{-4}	4.91×10^{-5}
80	0.5	1.68×10^{-4}	5.59×10^{-5}
	1.0	1.72×10^{-4}	5.56×10^{-5}
	2.0	1.91×10^{-4}	5.51×10^{-5}
	4.0	2.10×10^{-4}	5.46×10^{-5}
	8.0	2.35×10^{-4}	5.42×10^{-5}

Table 2: Summary of Species Mass Transfer Coefficients for Series Configuration

Series Configuration			
Air Pressure (psi)	Bottom Air Flow Rate ($\frac{L}{min}$)	$K_{O_2}(\frac{mol}{min * Pa})$	$K_{N_2}(\frac{mol}{min * Pa})$
20	0.5	1.51×10^{-4}	4.34×10^{-5}
	1.0	1.78×10^{-4}	4.23×10^{-5}
	2.0	1.83×10^{-4}	4.19×10^{-5}
	4.0	1.79×10^{-4}	4.10×10^{-5}
	8.0	1.91×10^{-4}	3.94×10^{-5}
40	0.5	2.14×10^{-4}	5.00×10^{-5}
	1.0	2.09×10^{-4}	5.02×10^{-5}
	2.0	2.20×10^{-4}	5.03×10^{-5}
	4.0	2.37×10^{-4}	4.93×10^{-5}
	8.0	2.45×10^{-4}	4.84×10^{-5}
80	0.5	2.07×10^{-4}	5.60×10^{-5}
	1.0	2.11×10^{-4}	5.56×10^{-5}
	2.0	2.21×10^{-4}	5.57×10^{-5}
	4.0	2.39×10^{-4}	5.51×10^{-5}
	8.0	2.60×10^{-4}	5.46×10^{-5}

Overall, a higher air flow rate provides a better separation. And the series configuration is more effective in separating the dry air than the parallel counterpart.

2. Introduction

The objective of this experiment was to pass dry air through two hollow fiber polymeric membrane modules and determine the mass transfer coefficients of O_2 and N_2 for parallel and series configurations. Experiments involving separating air using a membrane are important for chemical engineers to understand, as the process is widely used in industry. For instance, Linde Engineering utilizes membrane air separation to process contaminated air without the need for solvents while minimizing carbon footprint at the same time [1]. In addition, membrane air separation technology is also widely used in the pharmaceutical industry for drug separation and purification [2]. In this experiment, two membrane separator modules were used to separate oxygen and nitrogen from dry air, a transducer was used to monitor the pressure of incoming dry air, four 3-way directional

valves were used to direct dry air throughout the system, a manifold was used to distribute dry air throughout the system, two oxygen analyzers were used to measure oxygen amount at the separators' outlet, and two volumetric flowmeters were used to monitor the flow rate of incoming dry air. The experiment began by setting up the system's 3-way directional valves according to the instructions in the lab manual for a parallel configuration [3], and the two O_2 digital analyzers were also calibrated using the provided manual. The valve on top of the gas cylinder was then opened, allowing the dry air to flow towards the setup, which was monitored with a two-stage pressure regulator. Using the digital display of the transducer, the air pressure was set to 20 psi. The air flow rate was then set up to be $0.5 \frac{L}{min}$ using the needle valve in front of the volumetric flowmeter. As the air exits the modules, readings on the two digital O_2 analyzers were allowed to stabilize before being recorded. The remaining air flow rates were then performed using an identical procedure used for $0.5 \frac{L}{min}$. After all air flow rates have been performed, the subsequent air pressures and their corresponding flow rates were then adjusted accordingly and experimented using an identical procedure described before.

3. Theory

From the lab manual and [4], membrane gas separation is a diffusional mass-transfer process where gas molecules permeate through a nonporous polymeric membrane. The rate at which these molecules permeate is dependent on their solubility & diffusivity. The solution-diffusion model accounts for this transport, in which separation is governed by differences in each gas's solubility and diffusion through the polymer matrix. Here, the gas molecules adsorb onto the high-pressure side of the membrane first, then dissolve within the polymer phase, after which they diffuse across the membrane thickness and finally desorb at the low-pressure end. The steady state flux of a gas through the membrane is described as both Fick's & Henry's Law, respectively. Together, they give us the permeation equation: $J_A = \frac{q_A}{Z} (p_{A,feed} - p_{A,permeate})$, where q_A is the permeability coefficient and is given by: $q_A = D_A S_A$. The ratio between two gases defines the ideal selectivity: $\alpha = \frac{q_A}{q_B}$, which helps visualize the membrane's ability to differentiate between gas species (e.g., oxygen and nitrogen). Experimentally, separation is defined by the

separation factor: $\alpha' = \frac{(\frac{y_A}{y_B})_{permeate}}{(\frac{y_A}{y_B})_{retentate}}$, and a higher value indicates a more effective

separation. The overall mass transfer coefficient can be defined to help represent the combined effect of diffusion & resistance to mass transfer in the membrane:

$F_A = K_A (p_{A,feed} - p_{A,permeate})$. Additionally, the use of recovery & stage cut, given by:

$Stage\ Cut = \frac{Q_p}{Q_f}$, helps further characterize the membrane's performance. The recovery

measures the target component recovered and the stage cut gives the proportion of feed gas that permeates the membrane. Overall, the permeability, selectivity, and transport coefficients obtained experimentally help define the membrane's performance under the varying feed pressures, flow rates & configurations. A table summarizing equation variables and their corresponding units is provided below.

Table 3: Summary of Equation Variables and Their Units

Variable	Units	Description
J_A	$\frac{mol}{m^2 \cdot s}$	Steady-state flux of gas A through the membrane
q_A	$\frac{mol \cdot m}{m^2 \cdot s \cdot Pa}$	Permeability coefficient of gas A, combining diffusivity & solubility
Z	m	Membrane thickness
$p_{A,feed}$	Pa	Partial pressure of gas A on the feed (high-pressure) end
$p_{A,feed}$	Pa	Partial pressure of gas A on the permeate (low-pressure) end
D_A	$\frac{m^2}{s}$	Diffusion coefficient of gas in the membrane polymer
S_A	$\frac{mol}{m^3 \cdot s}$	Solubility coefficient of gas A in the membrane polymer
α	-	Ideal selectivity; ratio of $\frac{q_A}{q_B}$ for two gases
α'	-	Experimental separation factor; ratio of mole fraction ratios between permeate and retentate
y_A, y_B	-	Mole fractions of components A & B in the gas mixture
F_A	$\frac{mol}{m^2 \cdot s}$	Overall mass transfer flux of gas A across the membrane

K_A	$\frac{\text{mol}}{\text{min} \cdot \text{Pa}}$	Overall mass transfer coefficient for gas A, accounting for total resistance
Q_p	$\frac{\text{mol}}{\text{s}}$, or $\frac{\text{m}^3}{\text{s}}$	Permeate flow through the membrane
Q_f	$\frac{\text{mol}}{\text{s}}$, or $\frac{\text{m}^3}{\text{s}}$	Feed gas flow rate entering the membrane module
Stage Cut	-	Fraction of feed gas that permeates through the membrane (Q_p/Q_f)
Recovery	-, or %	Fraction of target component recovered in the permeate stream

4. Results & Data Analysis

Provided below are charts that plot the overall mass transfer coefficients of O_2 and N_2 at different pressure and air flow rates for parallel and series configurations, respectively

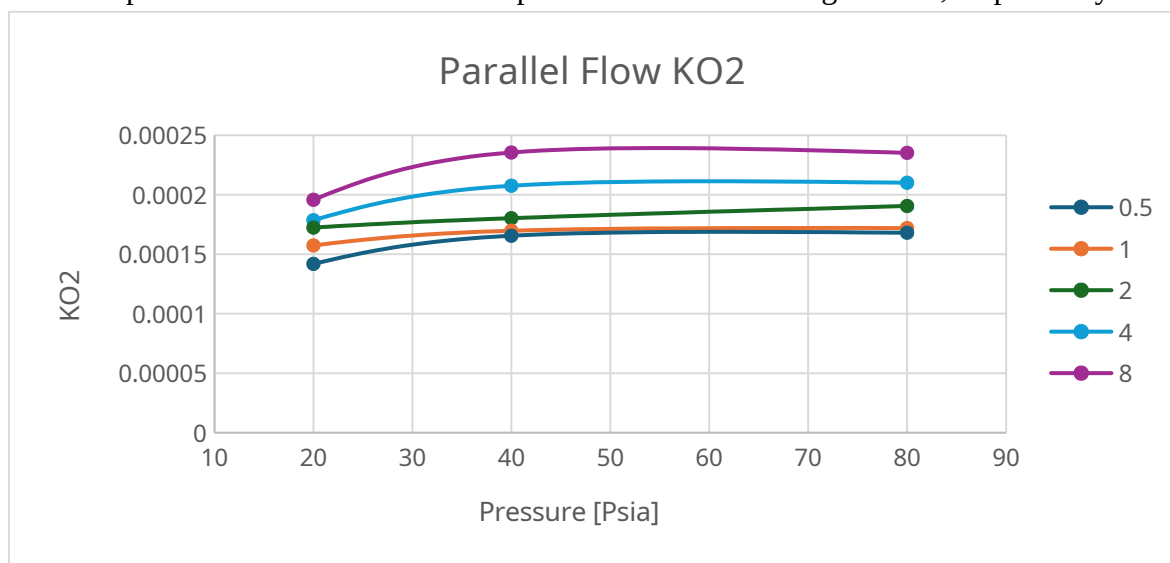


Figure 1: KO2 vs Pressure at Different Air Flow Rates for Parallel

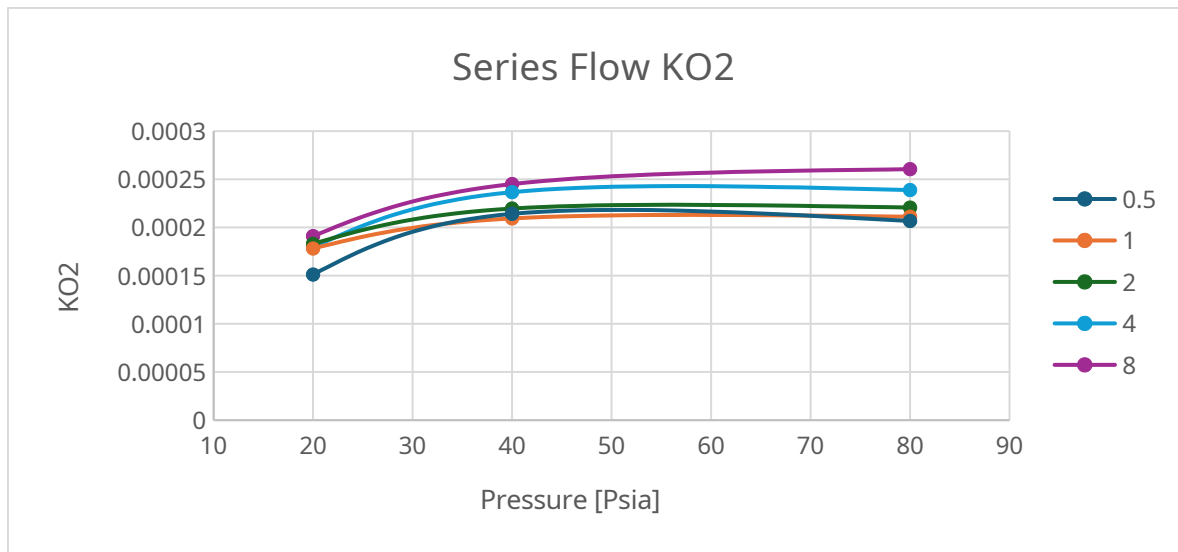


Figure 2: KO₂ vs Pressure at Different Air Flow Rates for Series

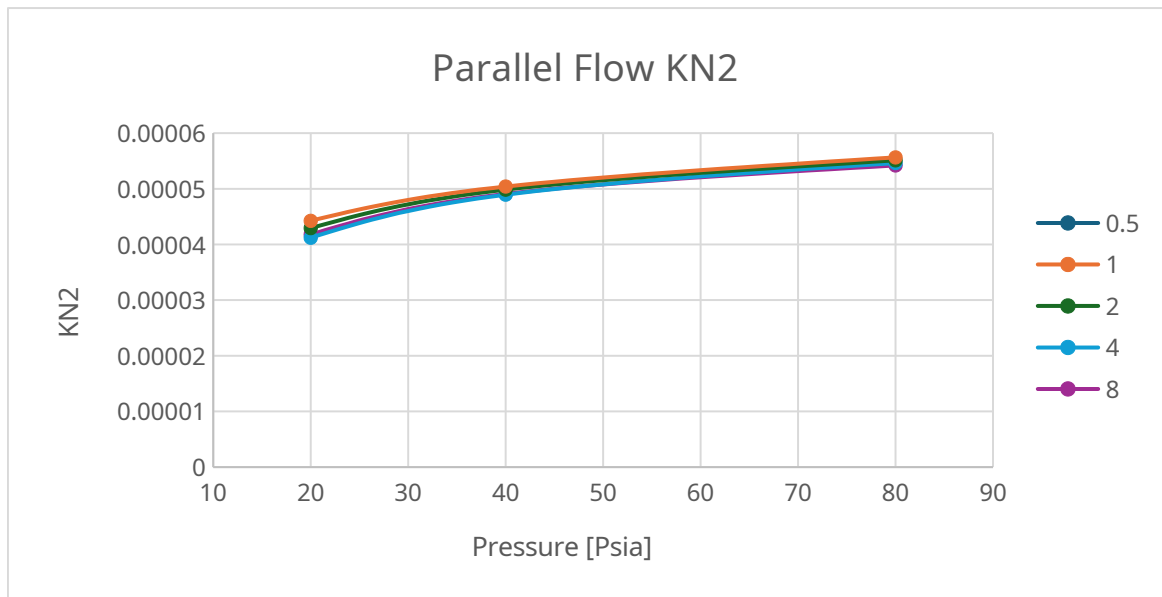


Figure 3: KN₂ vs Pressure at Different Air Flow Rates for Parallel

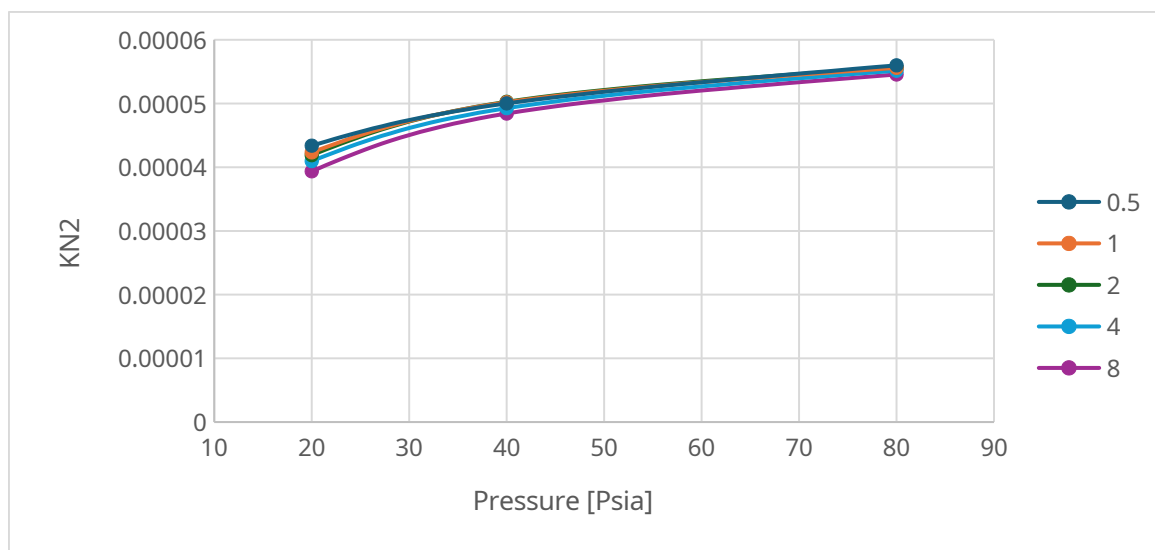


Figure 4: KN_2 vs Pressure at Different Air Flow Rates for Series

In the parallel and series configurations the overall mass transfer coefficient increased during the initial pressure rise from 20 to 40 psig and then has negligible change between 40 and 80 psig. This behavior reflects the fact that the mass transfer across the membrane is tied to the pressure drop across the membrane. At the lower pressures the system has limited oxygen flux across the membrane due to the pressure drop, the system becomes flux limited above 40 psig. Among all variable combinations the highest mass transfer occurs in the 8L/min 40-80 psig region showing that at moderate to high pressure and high flow are the best.

For the nitrogen mass transfer coefficient, there are negligible differences between pressures and flows but a positive correlation between overall nitrogen transport and pressure applied to the system, as the membranes in question for this experiment are porous only to oxygen any nitrogen that can cross the boundary is due to the pressure differential between shell and tube and not the affinity for the membrane material itself.

The difference between parallel and series system configurations is mostly negligible with the series system achieving slightly better mass transfer in a tighter grouping between flow rate variation, this is likely due to the gas effectively having double the membrane surface area to flux through, compared to the flow being split between the two membranes in parallel.

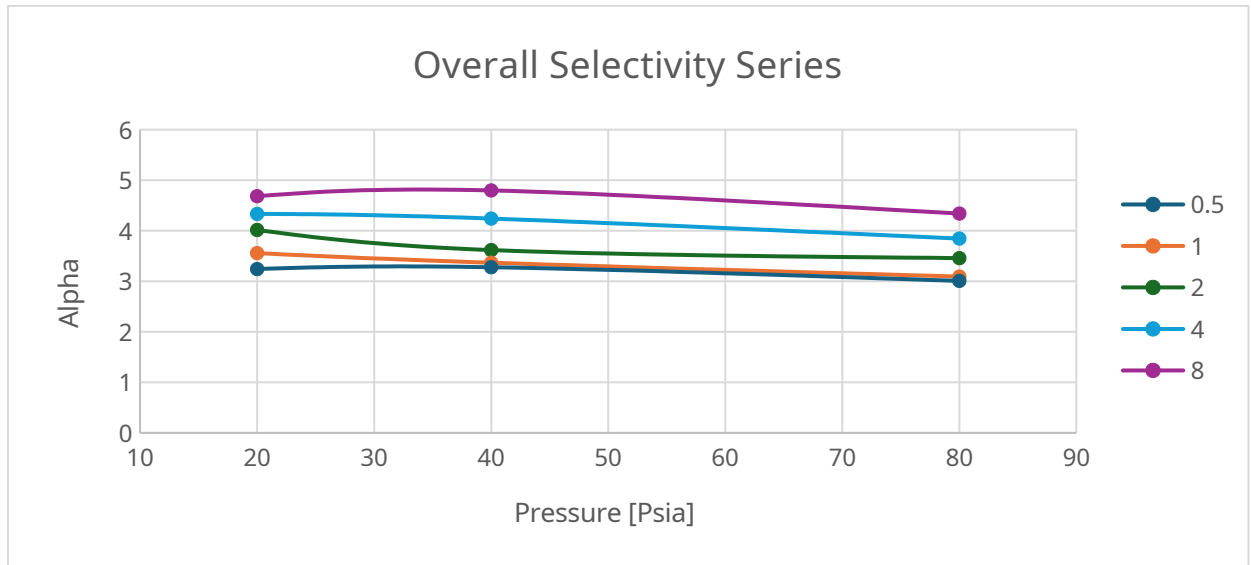


Figure 5: Overall Selectivity For Series Flow vs Pressure at Different Air Flow Rates

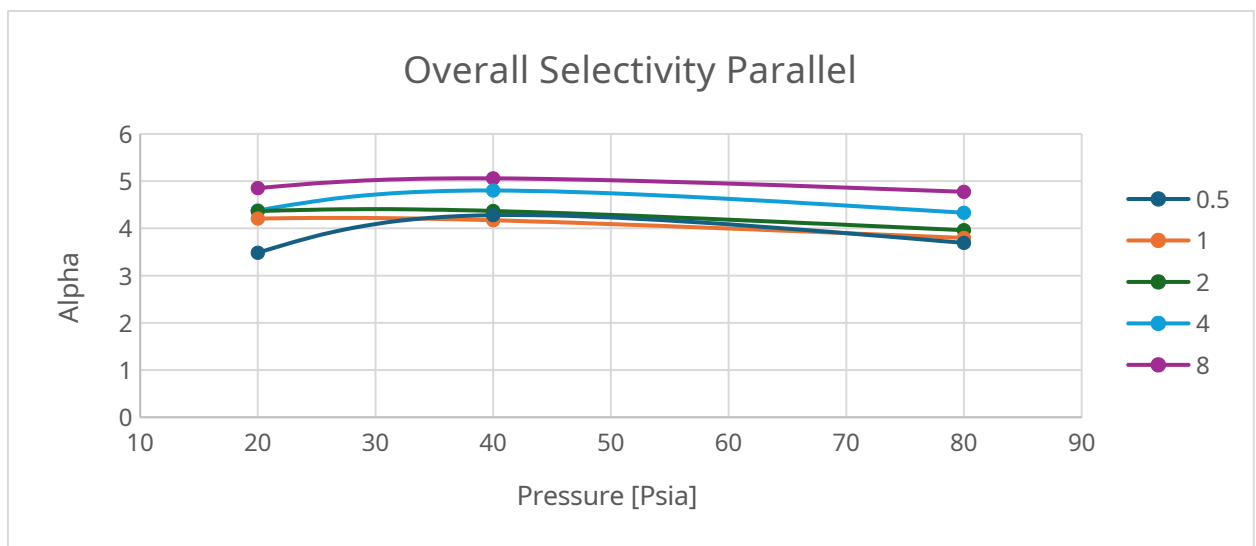


Figure 6: Overall Selectivity For Parallel Flow vs Pressure at Different Air Flow Rates

Overall selectivity ranged from 3 to 5 overall all variables tested. In both configurations, parallel and series, selectivity peaked at 40 psig and 8L in both system configurations. This peak at 40 psig instead of 80 psig is due to the pressure vs membrane affinity. At 40 psig, the system is likely not dominated by one specific force, pressure differential, or membrane affinity, this causes the selectivity to peak as the nitrogen present in the system is restricted by affinity and the pressure differential which is why at both high and low pressure selectivity decreases as the driving forces that the nitrogen is using to cross the boundary are exacerbated.

5. Discussion

The purpose of this experiment was to investigate how pressure and air flow rate influence the mass transfer coefficients for oxygen and nitrogen across membranes in both parallel and series configurations. The trends in Figures 1-4 align closely with the solution diffusion model. According to the theory, permeation is driven by gas solubility and diffusivity depending on the membrane material. So, gases with smaller molecular sizes, such as oxygen, are expected to have higher permeability than nitrogen.

In both the series and parallel configurations, the KO_2 values increased sharply between 20 and 40 psia and increased more gradually from 40 to 80 psia. This behavior suggests that the initial pressure difference was the main driving force, and the low pressures were ideal for the transport of oxygen within the membrane. Around 40 psia, the system seems to reach a limit, and a further increase in pressure has little to no effect on improving the oxygen transport. It's also been observed in the transport of oxygen that increasing the flow rate of air also increases the mass transfer coefficients of O_2 . Unlike the pressure, increasing the flow rates of air seems to steadily increase the transport of O_2 within the membrane, indicating that flow rates of air have a more direct impact on O_2 transport than air pressure, agreeing with findings in [5]. This could be because by increasing the air pressure, more air molecules are introduced into the system, creating a "clog" of molecules as they travel through the membranes. However, increasing the air flow rates provides additional kinetic energy for the molecules, allowing them to diffuse through the membrane more easily.

Unlike oxygen, the nitrogen mass transfer coefficient remained almost constant with increasing pressure and flow rate in series and parallel configurations, suggesting that the nitrogen transport is strongly limited by its diffusivity within the membrane. In other words, the nitrogen molecules don't have a strong affinity towards the membrane like oxygen molecules do. Because of this, the flux of nitrogen is far less sensitive to changes in pressure, which is consistent with the solution diffusion model. By observing the graph trends, it confirms that the increase in pressure effect on nitrogen permeation is very minimal.

6. References

- [1] <https://www.linde-engineering.com/products-and-services/process-plants/adsorption-and-membrane-plants/membrane-plants>
- [2] <https://pubmed.ncbi.nlm.nih.gov/38248714/>
- [3] Membrane Air Separation Lab Manual
- [4] <https://www.alfalaval.us/products/separation/membranes/what-is-membrane-filtration/>
- [5] <https://www.sciencedirect.com/science/article/abs/pii/S0043135499003231>

7. Appendix I: Data Tabulations/Graphs

Parallel					Series				
Air pressure (psi)	Top Flow rate (L/min)	Bottom flowrate (L/min)	Top oxygen (%)	Bottom oxygen (%)	Air pressure (psi)	Top Flow rate (L/min)	Bottom flowrate (L/min)	Top oxygen (%)	Bottom oxygen (%)
20	1.25	0.48	27.3	14.2	20	1.24	0.5	26.6	12.5
20	1.29	1	28.6	15	20	1.26	1	29.1	14
20	1.29	2	30.5	16.3	20	1.27	2	30.6	15.6
20	1.27	4	32.3	17.9	20	1.26	4	32	17.4
20	1.32	8	33.9	19.2	20	1.25	8	34.1	19.1
40	2.92	0.5	28.6	10.4	40	3	0.5	29.7	8.2
40	2.92	1	28.5	9.9	40	3	1	29.6	8.4
40	2.96	2	30.4	11.2	40	3.06	2	31	9.4
40	3.06	4	33.9	13.1	40	3.14	4	34.5	12.1
40	3.23	8	37.4	15.4	40	3.22	8	37.8	15.1
80	6.32	0.5	27.1	6.1	80	6.53	0.5	28.2	4.1
80	6.37	1	28.2	6.9	80	6.52	1	28.5	4
80	6.46	2	29.5	6.5	80	6.63	2	29.7	4.6
80	6.67	4	32.6	8.1	80	6.84	4	32.9	6.5
80	6.99	8	36.6	10.3	80	7.16	8	37.2	9.3

Figure 5: Raw Data of the Experiment

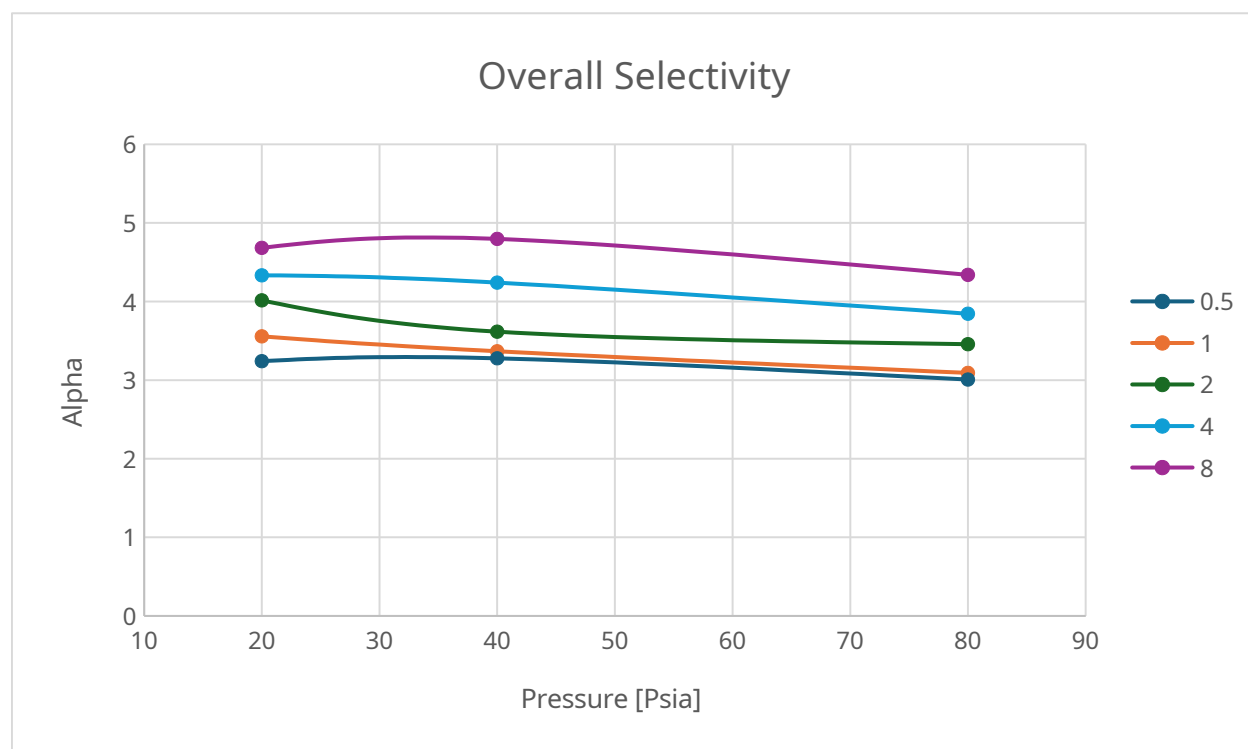


Figure 6: Overall Selectivity vs Pressure at Different Air Flow Rates

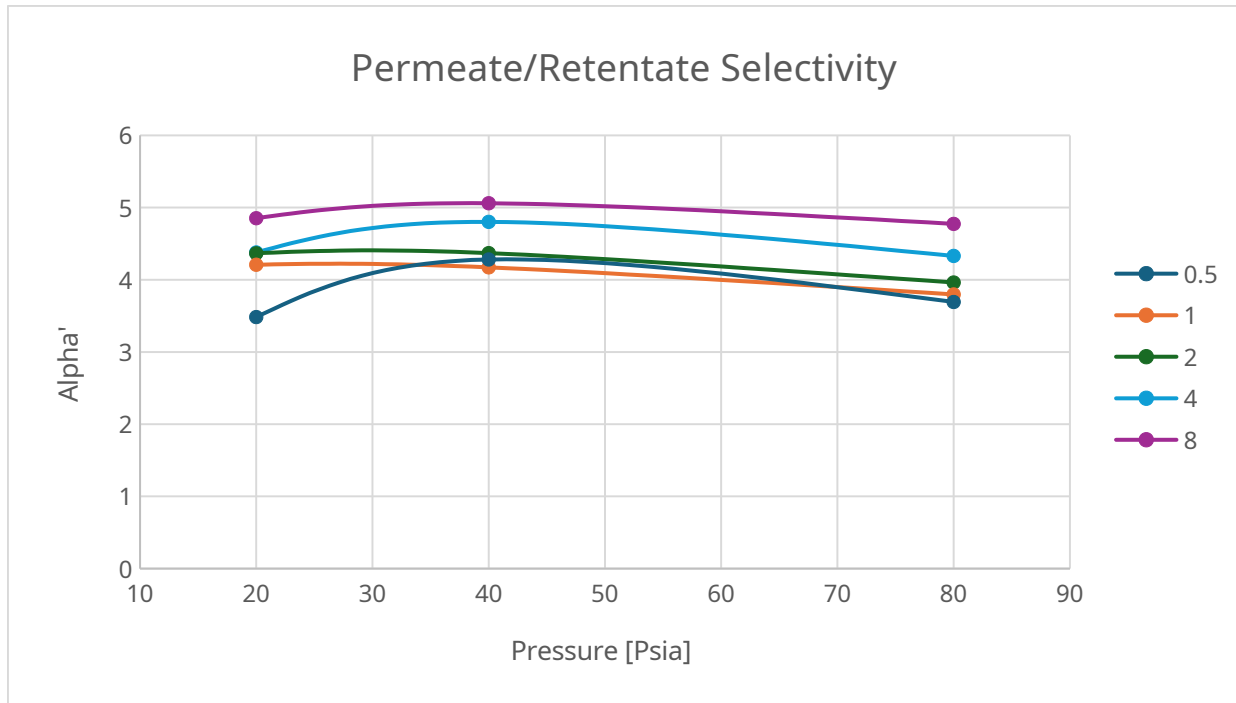


Figure 7: Permeate Selectivity vs Pressure at Different Air Flow Rates

8. Appendix II: Error Analysis

There can be some experimental errors that may have influenced the observed data. Minor pressure fluctuations during operation or potential leaks in the tubing or fittings could have changed the actual pressure drop across the membrane. Another error could be the assumption of steady state within the system, as the behavior could possibly not have been perfect. To minimize these in the future, digital flow meters should be used, and leak testing should be conducted before each run.

9. Appendix III: Sample Calculations

Parallel Calculations:

To do these calculations, the first set of data will be used, which is the following:

Air Pressure: 20 Psi

Top Flow rate: 1.25 L/min

Bottom Flow Rate: 0.48 L/min

Top Oxygen: 27.3 %

Bottom Oxygen: 14.2%

To start, it is necessary to convert units from psi to torr:

$$20 \text{ psi} + 14.7 \frac{\text{psi} * 760 \text{ Torr}}{14.7 \text{ psi}} = 1794.01 \text{ Torr}$$

Now calculating the Air Flow of the Top unit:

$$\frac{(1.25 \frac{\text{L}}{\text{min}} * 1000)}{22400} = 0.0558 \frac{\text{mol}}{\text{min}}$$

Based on the gas cylinder that is being used for the lab experiment it is safe to assume that the air being fed through the pipes is comprised of 20.9% oxygen and 79.1% Nitrogen.

This gives the following mol fractions:

$$X_{O_2} = 0.209 \wedge X_{N_2} = 0.791$$

Calculating the Air Flow for the Bottom Unit:

$$\frac{0.48 \frac{\text{L}}{\text{min}} * 1000}{22400} = 0.0412 \frac{\text{mol}}{\text{min}}$$

Next, calculate the mol fraction of Bottom Oxygen X_{1B}

$$\frac{\% \text{ Bottom Oxygen}}{100} = \frac{14.2}{100} = 0.142$$

Also, calculate the mol fraction of Top Oxygen X_{1S}

$$\frac{\% \text{ Top Oxygen}}{100} = \frac{27.3}{100} = 0.273$$

Can now find partial pressure for the mol fractions of X_{O_2} , X_{1B} , X_{1S}

$$\text{Total Pressure} * X_{O_2} = 1794.01 \text{ torr} * 0.209 = 374.95 \text{ torr}$$

$$\text{Total Pressure} * X_{1B} = 1794.01 \text{ torr} * 0.142 = 254.75 \text{ torr}$$

$$\text{Total Pressure} * X_{1S} = 1794.01 \text{ torr} * 0.273 = 207.48 \text{ torr}$$

After finding the Partial Pressures need to find the average between P_{O_2} , P_{1B} in order to calculate K_{O_2} later. With the average being $P_{1t} = 314.84 \text{ torr}$

Now calculating the molar fractions for Nitrogen. Since The mol fractions of Oxygen is known and knowing the mixture must add up to one.

$$X_{2B} = 1 - X_{1B} = 1 - 0.142 = 0.858$$

$$X_{2S} = 1 - X_{1S} = 1 - 0.273 = 0.727$$

Calculate the partial pressure for each of the mol fractions again.

$$\text{Total Pressure} * X_{N_2} = 1794.01 \text{ torr} * 0.791 = 1419.06 \text{ torr}$$

$$\text{Total Pressure} * X_{2B} = 1794.01 \text{ torr} * 0.858 = 1539.26 \text{ torr}$$

$$\text{Total Pressure} * X_{2S} = 1794.01 \text{ torr} * 0.727 = 552.52 \text{ torr}$$

After finding the Partial Pressures need to find the average between P_{N_2} , P_{2B} in order to calculate KN_2 later. With the average being $P_{2t} = 1479.16 \text{ torr}$

From here it is now possible to calculate for $KO_2 \wedge KN_2$ with the following equation:

$$KO_2 = \frac{X_{1S} * \text{Air Flow Rate}}{(P_{1t} - P_{1s})} = \frac{0.273 * 0.0558 \frac{\text{mol}}{\text{min}}}{314.84 \text{ torr} - 207.84 \text{ torr}} = 0.000142 \frac{\text{mol}}{\text{min} * \text{torr}}$$

$$KN_2 = \frac{X_{2S} * \text{Top Air Flow Rate}}{(P_{2t} - P_{2s})} = \frac{0.791 * 0.0558 \frac{\text{mol}}{\text{min}}}{1479.16 \text{ torr} - 552.52 \text{ torr}} = 4.38 * 10^{-5} \frac{\text{mol}}{\text{min} * \text{torr}}$$

Now to find the permeability:

$$\text{Permeability} = \frac{KO_2}{KN_2} = 3.24 \frac{\text{mol}}{\text{sec}}$$

Calculate the retentate:

$$\text{Retentate} = \frac{X_{1S} / X_{2S}}{X_{1B} / X_{2B}} = \frac{\frac{0.273}{0.727}}{\frac{0.142}{0.858}} = 2.27$$

Calculate the feed:

$$\text{Feed} = \frac{X_{1S} * X_{2S}}{X_{1a} / X_{2a}} = \frac{\frac{0.273}{0.727}}{\frac{0.209}{0.727}} = 1.42 \text{ mols}$$

Calculate the Stage cut:

$$\frac{\text{Top Air Flow}}{(\text{Top Air Flow} + \text{Bottom Air Flow})} = \frac{0.0558 \frac{\text{mol}}{\text{min}}}{(0.0558 \frac{\text{mol}}{\text{min}} + 0.0241 \frac{\text{mol}}{\text{min}})} = 0.723$$

Calculations for pressures at 40 and 80 psi, as well as the flow when it is in series, followed an identical procedure to the calculations provided above.

10. Appendix IV: Individual Team Contributions

ASSIGNMENT	TIME (HOURS)	DESCRIPTION
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Operator (Both Lab Days)	5	All Team members
Pre-Lab Editing	2	All Team members
Experimental Objective (Prelab)	1	Rishabh Patel
Apparatus (prelab)	1	Khoa Nguyen
Materials & Supplies (prelab)	1	Khoa Nguyen
Procedure (prelab)	1	Alexis Guillen
Project Safety Evaluation (prelab)	1	Kyle Hankosky
Final Lab Editing	1	All Team members
Executive Summary (final lab)	2	Khoa Nguyen & Rishabh Patel
Introduction (Final lab)	1	Khoa Nguyen & Rishabh Patel
Theory (Final Lab)	4	Alexis Guillen & Khoa Nguyen
Results (final lab)	2	Kyle Hankosky & Alexis Guillen
Discussion (final lab)	2	Khoa Nguyen
References (final and/or prelab)	1	Rishabh Patel
Data Tabulation/Graphs (final)	2	Kyle Hankosky & Alexis Guillen
Error Analysis (final lab)	1	Kyle Hankosky
Sample Calculations (final Lab)	2	Kyle Hankosky & Alexis Guillen

PowerPoint Presentation	5	All Team members
Total	35 Hours	